Aryana Haghjoo

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Aryana Haghjoo is an early-career astrophysicist with a strong foundation in both theoretical and data-driven research. She holds an M.Sc. in Physics from McGill University, where she specialized in computational global 21cm cosmology, and is currently pursuing a Ph.D. in Astronomy at the University of California, Riverside. Her recent research leverages deep learning to super-resolve low-resolution galaxy spectra.

Education

Ph.D. Astronomy, University of California, Riverside (UCR) — *Sept 2025 – Sept 2028 (Expected)*
GPA: 3.7/4 — *Riverside, California, USA*

M.Sc. Astronomy, UCR — *Sept 2024 – Jun 2026*
GPA: 3.7/4 — *Riverside, California, USA*

M.Sc. Physics, McGill University — *Sept 2021 – Dec 2023*
GPA: 3.5/4 — *Montreal, Quebec, Canada*

Thesis: Parameter Estimation of the Global 21cm Signal

B.Sc. Physics, Sharif University of Technology (SUT) — *Sept 2018 – Jul 2021*
GPA: 18.25/20 — *Tehran, Iran*

Publications

14 publications, 106 citations, h-index: 6 (NASA ADS, Oct 2026).

First-Author Manuscripts

- **Haghjoo, A.**, Hemmati, S., Mobasher, B., Chartab, N., de la Vega, A., Eifler, T., Ramey, E., Nayyeri, H., & Sattari, Z. *Learning to See Sharper: A Physics-Informed Artificial Intelligence Framework for Super-Resolving Galaxy Spectra.* Under review at the *Astrophysical Journal*; arXiv:2603.18357.
- **Haghjoo, A.**, Hemmati, S., & Mobasher, B. *Sharper at What Cost? Benchmarking Deep Learning Against Classical Deconvolution for Galaxy Spectral Super-Resolution.* Under review at the *Astrophysical Journal*.
- **Haghjoo, A.**, Hemmati, S., Mobasher, B., Chartab, N., & Sattari, Z. *Spectral Super-Resolution for Roman Grism Spectroscopy.* In preparation for submission to the *Astrophysical Journal*.
- **Haghjoo, A.**, Mobasher, B., & D'Aloisio, A. *Connecting Observational Spectra to the Bubble Size Distribution at the Epoch of Reionization with Deep Learning.* In preparation for submission to the *Astrophysical Journal*.

Selected Co-Author Publications

- Taamoli, S., Mobasher, B., Hatamnia, H., **Haghjoo, A.**, et al. *The Role of the Cosmic Web in Galaxy Evolution Across* $0.4 < z < 4$. *arXiv preprint* (2026).
- Petric, A., Noirot, G., Alberts, S., **Haghjoo, A.**, et al. *Surveying the Universe in 4D: Beating Cosmic Variance with Wide-Field Slitless Spectroscopy from HST, JWST, Euclid, Roman, and Beyond. arXiv preprint* (2026).
- Zhang, Y., Ding, X., Yang, L., **Haghjoo, A.**, et al. *Extended Components of Little Red Dots in the Rest-Frame Optical. Nature Astronomy* (2026).
- Aparicio, V., Jones, D. O., Hoogendam, W. B., **Haghjoo, A.**, et al. *Analysis of a Type II Supernova Candidate at* $z = 3.19$ *from JWST's COSMOS-Web Survey. Astrophysical Journal* (2026).
- Abedini, F., Gozaliasl, G., Zonoozi, A. H., **Haghjoo, A.**, et al. *COSMOS-Web: Estimating Physical Parameters of Galaxies Using Self-Organizing Maps. Monthly Notices of the Royal Astronomical Society* (2026).
- Hatamnia, H., Mobasher, B., Taamoli, S., **Haghjoo, A.**, et al. *Large-scale Structure in COSMOS-Web: Tracing Galaxy Evolution in the Cosmic Web up to* $z \sim 7$ *with the Largest JWST Survey. Astrophysical Journal* (2026).
- Shuntov, M., Ilbert, O., Lagos, C. del P., **Haghjoo, A.**, et al. *The Stellar Mass Function of Quiescent and Star-Forming Galaxies and its Dependence on Morphology in COSMOS-Web. Astronomy & Astrophysics* (2026).

- Yu, S.-Y., Ho, L. C., Tsukui, T., **Haghjoo, A.**, et al. *Through Thick and Thin: The Cosmic Evolution of Disk Scale Height. Astrophysical Journal Supplement Series* (2026).
- Hemmati, S., Krick, J., Stern, D., **Haghjoo, A.**, et al. *Reducing the Dimensions of Active Galactic Nuclei Light-Curve Manifolds. Astrophysical Journal* (2026).
- Toni, G., Maturi, M., Castignani, G., **Haghjoo, A.**, et al. *COSMOS-Web Galaxy Groups: Evolution of the Red Sequence and Quiescent Galaxy Fraction. Astronomy & Astrophysics* (2026).
- Fujimoto, S., Faisst, A. L., Tsujita, A., **Haghjoo, A.**, et al. *The ALPINE-CRISTAL-JWST Survey: NIRSpec IFU Data Processing and Spatially Resolved Views of Chemical Enrichment in Normal Galaxies at $z = 4$–6. arXiv preprint* (2025).
- Gozaliasl, G., Yang, L., Kartaltepe, J. S., **Haghjoo, A.**, et al. *Brightest Group Galaxies in COSMOS-Web: Evolution of the Size–Mass Relation Since $z = 3.7$. Astronomy & Astrophysics* (2025).
- Asadi, V., Chartab, N., Zonoozi, A. H., **Haghjoo, A.**, et al. *Machine Learning Classification of COSMOS2020 Galaxies: Quiescent versus Star-Forming. Astrophysical Journal* (2025).

Research Experiences and Projects

Super-Resolving Low-Resolution Galaxy Spectra with Deep Learning *Dec 2024 – Present*
Under the supervision of Bahram Mobasher and Shoubaneh Hemmati *UCR*

- Developed a generative model to enhance spectral resolution of PRISM data by a factor of 10 using data from the JADES program
- Trained and validated the model with a physics-informed loss function targeting emission line accuracy and flux ratios (e.g., Hα/Hβ), achieving improved deblending of close spectral features
- Demonstrated the potential of the method to increase the resolution of deep spectral data from future instruments (e.g., Nancy Grace Roman Space Telescope)
- Currently extending the model to a multi-modal foundation framework that integrates photometric and spectroscopic data, enabling cross-modal inference

Investigating the Effects of Non-Standard Physics on the Global 21cm Signal *Sept 2021 – Dec 2023*
Under the supervision of Jonathan Sievers and Oscar Hernandez *McGill University*

- Investigated the potential of the global 21cm signal to reveal non-standard physics
- Developed a pipeline using Markov Chain Monte Carlo (MCMC) and Levenberg-Marquardt (LM) algorithms to estimate the astrophysical parameters of the global signal
- Employed Accelerated Reionization Era Simulations (ARES) to generate theoretical models
- Validated the approach using data from Experiment to Detect the Global EoR Signature (EDGES)
- Results will aid in constraining future models and defining theoretical limits for upcoming experiments

Honours and Awards

USC AI4SE Hackathon: Most Advanced Technology *2026*
UCR Dean's Fellowship *2024*
McGill Faculty of Science Graduate Award *2023*
McGill Graduate Excellence Award *2021*
Trottier Space Institute at McGill Graduate Fellowship *2021*
National University Entrance Examinations: Top 0.5% Among 180K Candidates *2017*

Presentations

From JWST to Roman: A Physics-Informed AI Super-Resolution Framework for Grism Spectroscopy *Sept 2026*
Verbal presentation at GalFRESCA 2026 (Galaxy Formation & Evolution in Southern California) *UC Irvine*

From JWST to Roman: A Physics-Informed AI Super-Resolution Framework for Grism Spectroscopy *Sept 2026*
Verbal presentation at Open SkAI 2026, SkAI Institute *Chicago, IL*

From JWST to Roman: A Physics-Informed AI Super-Resolution Framework for Grism Spectroscopy *Aug 2026*
Verbal presentation at the Surveying the Universe in 4D Workshop (virtual) *Space Telescope Science Institute*

Learning to See Sharper: A Physics-Informed AI Framework for Spectral Super-Resolution *Aug 2026*
Poster presentation at the Sierra Conference 2026 (UC astronomy graduate students) *Yosemite National Park*

From JWST to Roman: An AI Super-Resolution Framework for Grism Spectroscopy *Jul 2026*
Verbal presentation at the Shifting Landscapes in Astrophysics *IPAC, Caltech*

Deep Learning for Astronomical Data Enhancement *Jun 2026*
Verbal presentation at the 248th AAS Meeting *Pasadena, CA*

Learning to See Sharper: A Physics-Informed AI Framework for Spectral Super-Resolution *Jun 2026*
Poster presentation at the 2026 Conference on Physics and AI (PAI26) *Stanford University*

Learning to See Sharper: A Physics-Informed AI Framework for Spectral Super-Resolution *Apr 2026*
Verbal presentation at the AstroAI Lunch Talks (Invited) *CfA, Harvard and Smithsonian*

Learning to See Sharper: A Physics-Informed AI Framework for Spectral Super-Resolution *Mar 2026*
Verbal presentation at the Euclid US Tag-up Meeting (Invited) *Euclid Consortium*

Super-Resolving Low-Resolution Galaxy Spectra *Jan 2026*
Poster presentation at the Arizona Winter School on Cosmology in the Roman ST Era *University of Arizona*

Super-Resolving Low-Resolution Galaxy Spectra *Oct 2025*
Verbal presentation at the IR Spectroscopy from Space *IPAC, Caltech*

Super-Resolving Low-Resolution Galaxy Spectra *Aug 2025*
Verbal presentation at the Greater IPAC Science Symposium *IPAC, Caltech*

Super-Resolving Low-Resolution Galaxy Spectra *Jul 2025*
Poster presentation at the Cosmic Cartography with Roman *Space Telescope Science Institute*

Super-Resolving Low-Resolution Galaxy Spectra *Jul 2025*
Verbal presentation at the AstroAI Workshop *CfA, Harvard and Smithsonian*

Unraveling Non-Standard Physics Through the Global 21cm Signal *Jun 2023*
Verbal presentation at the SUT Cosmology Seminar *SUT*

Parameter Estimation of the Global 21cm Signal *Mar 2023*
Poster presentation at Cosmology on Safari 2023 *Hluhluwe, South Africa*

Clarifying the Effects of Non-Standard Physics on the Global 21cm Signal *Oct 2022*
Verbal presentation at the McGill Radio Lab Meetings *McGill University*

Effects of Non-Standard Physics on 21cm Signal *May 2022*
Verbal presentation at the McGill Early Universe Cosmology Meetings *McGill University*

Work Experience

Teaching Assistant *2025 – Present*
PHYS 243 (Applied Machine Learning); PHYS 247 (Data Science); PHYS 2C, 2B Discussions; PHYS 40A–C *UCR Labs*

Teaching Assistant *2022 – 2023*
PHYS 102 Lab; Galaxies and Stars *McGill University*

Teaching Assistant *2019 – 2020*
PHYS 103; Introduction to Cosmology (twice); Electromagnetism *SUT*

Skills and Languages

Programming	Python, MATLAB, LaTeX, Bash, C, C++, Quick Basic, Visual Basic
Platforms	GitHub, GitLab, SLURM
Languages	Persian (native), English (fluent, full professional proficiency)

Selected Courses

Astrophysics	Galaxy Formation and Evolution, Astrophysics of the Interstellar Medium, Stellar Structure and Evolution, Radiative Processes in Astrophysics, Astrophysical Fluid Dynamics (in progress), Observational Astrophysics
Cosmology & Gravity	Advanced Cosmology, General Relativity
Computation & Data	Machine Learning in Physics, Computational Physics, Modeling Statistical Phenomena, Complex Systems
Theoretical Physics	Quantum Field Theory, String Theory, Elementary Particles

Other Activities and Interests

Mentor at STEM High School: Assisting Senior Students on Their Projects	*Sept 2025 – Mar 2026*
Member, TEDxAveLorne Curation Team	*Jul 2023 – Dec 2023*
Representative, McGill Graduate Association of Physics Students (MGAPS)	*Jan 2023 – Dec 2023*
SUT Mentoring Program Volunteer: Senior-Freshman Buddy Program	*2020 – 2021*
Amateur Astronomer: Outreach Observing Nights	*2015 – 2020*

References

Bahram Mobasher	Distinguished Professor, UCR	mobasher@ucr.edu
Shoubaneh Hemmati	Research Scientist, Caltech	shemmati@ipac.caltech.edu
Jonathan Sievers	Associate Professor, McGill University	jonathan.sievers@mcgill.ca
Oscar Hernandez	Adjunct Professor, McGill University	oscarh@physics.mcgill.ca
Bahram Mashhoon	Emeritus Professor, University of Missouri	mashhoonb@missouri.edu
Shant Baghram	Associate Professor, SUT	baghram@sharif.edu